

TUAL DUMATUBUN, Indonesia

WMO: 978100

Lat: 5.6636S

Lon: 132.7361E

Elev: 12

StdP: 101.18

Time Zone: 9.00 (E09)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	22.6	19.9	14.6	24.7	20.6	15.2	25.1	11.4	26.3	10.3	26.4	0.4	150	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11	7.1	32.6	26.9	32.1	26.8	31.6	26.7	28.1	31.0	27.7	30.7	27.4	30.3	4.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.2	23.0	30.1	26.8	22.5	29.7	26.6	22.1	29.4	90.0	31.2	88.4	30.8	86.9	30.5	30.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.0	8.1	7.2	16.4	34.5	2.5	2.1	14.6	36.0	13.2	37.2	11.8	38.4	10.0	40.0
			16.9	28.7	2.0	0.8	15.5	29.3	14.3	29.7	13.2	30.1	11.8	30.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.3	27.5	27.3	27.3	27.4	27.5	26.9	26.5	26.3	27.0	27.8	28.0	27.5	
	DBStd	1.15	1.04	1.00	1.12	1.03	1.11	1.05	0.83	1.15	1.10	1.18	0.99	0.87	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6298	541	484	536	523	543	507	512	506	511	553	539	542	
	CDD18.3	3256	283	251	277	273	284	257	254	248	261	295	289	283	
	CDH23.3	32844	2939	2560	2811	2806	3033	2448	2199	2153	2529	3213	3205	2946	
	CDH26.7	9438	878	722	828	863	899	524	344	394	688	1161	1218	917	
Wind	WSAvg	3.0	3.0	2.8	2.3	2.0	2.7	3.7	4.4	4.4	3.5	2.6	1.8	2.1	
Precipitation	PrecAvg	2622	418	336	375	290	267	152	84	68	64	82	175	310	
	PrecMax	4150	630	700	580	577	714	396	278	251	232	169	328	550	
	PrecMin	1629	176	170	187	26	63	0	0	0	0	6	47	131	
	PrecStd	675	126	149	123	156	152	105	76	73	56	49	93	140	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	31.7	32.2	32.4	32.0	31.1	30.2	30.5	31.8	33.1	33.5	32.6	
		MCWB	27.0	26.6	27.2	27.2	27.3	26.6	25.9	25.5	26.2	27.1	26.9	26.9	
	2%	DB	31.4	31.1	31.5	31.7	31.3	30.5	29.6	29.8	31.1	32.3	32.7	31.7	
		MCWB	26.9	26.6	26.9	26.9	27.0	26.3	25.4	25.0	25.7	26.7	26.8	26.9	
	5%	DB	30.8	30.6	30.8	31.1	30.8	29.9	29.0	29.3	30.4	31.8	32.0	31.1	
		MCWB	26.8	26.5	26.7	26.7	26.8	26.0	25.1	24.8	25.4	26.4	26.7	26.8	
	10%	DB	30.2	30.0	30.2	30.5	30.2	29.3	28.5	28.8	29.8	31.1	31.4	30.4	
		MCWB	26.6	26.5	26.6	26.6	26.6	25.7	24.9	24.5	25.1	26.1	26.5	26.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.1	28.0	28.1	28.2	28.3	27.2	26.7	26.4	27.0	28.4	28.2	28.0
			MCDB	30.9	30.2	30.9	31.2	30.8	30.1	29.2	29.4	31.0	32.0	31.7	30.9
2%		WB	27.5	27.4	27.6	27.6	27.7	26.7	26.1	25.8	26.4	27.6	27.7	27.5	
		MCDB	30.3	29.8	30.4	30.5	30.4	29.6	28.7	28.8	30.1	31.3	31.2	30.4	
5%		WB	27.1	27.0	27.1	27.2	27.2	26.3	25.6	25.3	26.0	26.9	27.2	27.2	
		MCDB	29.9	29.5	29.9	30.1	30.0	29.1	28.2	28.2	29.5	30.6	30.7	30.0	
10%		WB	26.7	26.7	26.7	26.8	26.8	26.0	25.2	25.0	25.6	26.5	26.8	26.8	
		MCDB	29.3	29.1	29.4	29.6	29.5	28.7	27.6	27.8	28.8	30.0	30.2	29.6	
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.3	5.6	5.8	5.4	4.5	4.1	5.1	6.0	6.9	7.1	5.9	
		MCDBR	6.2	6.3	6.6	6.8	6.2	5.3	4.7	5.6	6.7	7.4	8.0	7.1	
		MCWBR	2.7	2.7	3.0	3.0	2.8	2.4	2.3	2.4	2.6	3.0	3.2	3.1	
	5% WB	MCDBR	5.7	5.5	6.1	6.3	5.9	5.0	4.5	5.0	6.3	6.7	7.2	6.4	
		MCWBR	2.7	2.8	3.2	3.2	3.2	2.6	2.4	2.5	2.7	3.0	3.4	3.2	
Clear-Sky Solar Irradiance	taub		0.415	0.414	0.402	0.405	0.434	0.440	0.429	0.433	0.446	0.467	0.449	0.419	
	taud		2.525	2.520	2.563	2.563	2.474	2.444	2.476	2.459	2.408	2.326	2.392	2.496	
	Ebn at Noon		923	923	921	892	839	819	836	855	868	864	885	916	
	Edh at Noon		111	112	106	102	107	108	106	112	122	134	126	114	
All-Sky Solar Radiation	RadAvg		4.76	5.08	5.07	5.00	4.73	4.35	4.64	5.29	5.81	6.14	5.94	4.99	
	RadStd		0.57	0.48	0.38	0.34	0.41	0.42	0.36	0.44	0.37	0.43	0.40	0.49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page